Year 12 Maths - Exponential models

Similar questions - Using logarithms to find exponential and power models

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A scientist believes the number of bacteria, N, in a culture t hours after the start is modelled by $N = ab^t$, where a and b are constants.

(a) Show that a graph of $\log_{10} N$ against t should be a straight line, and state its gradient and intercept in terms of a and b. **[3]**

(b) Her graph is a straight line with gradient 0.477 and vertical intercept 1.30. Find a and b to 2 significant figures. **[3]**

(c) Use the model to find the first whole number of hours after which there are more than one million bacteria. **[3]**

(d) State one reason why this prediction may be unreliable. **[1]**

2 Two quantities are related by $Q = aP^n$, where a and n are constants. When $P = 4$, $Q = 16$ and when $P = 16$, $Q = 128$.

(a) State what should be plotted on each axis to give a straight-line graph, and what the gradient and intercept represent. **[3]**

(b) Find the values of a and n. **[4]**

(c) Hence find the value of Q when $P = 100$. **[1]**

Question 1

(a) $\log_{10} N = \log_{10} a + t \log_{10} b$, which is linear in t. Gradient $\log_{10} b$, intercept $\log_{10} a$

(b) $b = 10^{0.477} = 3.0$ and $a = 10^{1.30} = 20$

(c) $20 \times 3^t > 10^6$, so $3^t > 5 \times 10^4$ and $t > \dfrac{\log_{10}(5 \times 10^4)}{\log_{10} 3} = 9.85$. After 10 hours

(d) E.g. the bacteria would run out of food or space, so exponential growth cannot continue indefinitely

Question 2

(a) Plot $\log Q$ against $\log P$ (since $\log Q = \log a + n \log P$). Gradient $= n$, intercept $= \log a$

(b) $\dfrac{128}{16} = \left(\dfrac{16}{4}\right)^n$, so $4^n = 8$ and $n = \frac{3}{2}$. $a = \dfrac{16}{4^{3/2}} = 2$

(c) $Q = 2 \times 100^{3/2} = 2000$